Band-1 Hadamard Order 9128 – Classical Paley-I (q=9127); IN SPK via Paley-I; Gap 2052 Still OPEN

H_9128 Paley-I PASS; Paley-II SKIP (q=4563=3^3*13^2 NOT prime power); Kronecker FAIL/SKIP (no admissible split); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-21

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-09-21_Band1_Hadamard_9104_PaleyI`; `2026-09-21_Band1_Hadamard_9124_PaleyII`; `2026-09-21_Band1_Hadamard_9092_PaleyI`.
Lab. `/workspace/experiments/band1/hadamard_9128_paleyI/`.
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Honesty stamp (non-negotiable).
This memo claims **finite construction of order 9128 only (verified max|HH^T-nI|=0), via classical Paley-I q=9127 (prime ===3 mod 4). 9128 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 9128: Paley-I **PASS**s ($q = 9127$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL** ($q = 4563 = 3^3 \cdot 13^2$ NOT prime power); Kronecker **FAIL**/SKIP this wave (note $9128 = 4 \cdot 2282$ but $2282 \not\equiv 0 \pmod 4$ – no Hadamard-admissible Kronecker split). **9128** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **9104+9124+9092+2220+8884+2236+2160+2140+8564 / 1064+2128 / ...** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

Abstract

Finite construction of order **9128**.

- H_{9128} = Paley-I ($q = 9127$) – classical SPK
- Paley-II for 9128: **SKIP** ($q = 4563 = 3^3 \cdot 13^2$ NOT prime power)
- Kronecker for 9128: **FAIL**/SKIP (no Hadamard-admissible split; prefer Paley-I)
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_9128.py`

Results. overall_pass=True; units 101/101; wall 619.0459s; constructed max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 9128 = q + 1$ with $q = 9127$ prime $\equiv 3 \pmod 4$ -> Paley-I eligible.
- Paley-II: $q = 4563 = 3^3 \cdot 13^2$ NOT prime power -> SKIP.
- Kronecker: $9128 = 4 \cdot 2282$ but 2282 NOT Hadamard-admissible; no admissible split – FAIL honest; prefer Paley-I.
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_9128 Paley-I	Paley-I q=9127	0	611.2812
H_9128 Paley-II	SKIP q=4563=3ˆ3*13ˆ2 NOT prime power	–	–
H_9128 Kronecker	FAIL/SKIP (4*2282; 2282 not admissible)	–	–

SPK closure note

Seeds include prior SPK through **9104** lineage (**9104**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; **1064+2128**; . . . ; **NOT** 8948/8644/8516/. . .) + verified **9128**. Next constructive gap **2052** (OPEN).

Quantity	Value
9128 in SPK	**True** (Paley-I q=9127)
Gap 2052	**OPEN**
clay_closed	false

Quantity	Value
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

q=2293->n=9172 triage – Paley-I q=9171 composite; Paley-II q=4585=5 7131 NOT prime power; Kronecker 4*2293 OR n=2052 standard SPK constructive gap – method-attack / lit OR q=2288->n=9152 Paley-I DIRECT q=9151 OR q=2297->n=9188 Paley-I DIRECT q=9187 OR method-attack SPK gap 2052 OR Track B / Object 1 meter await